Numerical Questions on Classical Cryptography

Q1. Encrypt the plaintext SECURITY using Caesar Cipher with key K = 3

Plaintext Letter	**S**	**E**	**C**	**U**	**R**	**I**	**T**	**Y**
Plaintext Value (P)	**18**	**4**	**2**	**20**	**17**	**8**	**19**	**24**
Key (K)	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**
P + K	**21**	**7**	**5**	**23**	**20**	**11**	**22**	**27**
(P + K) mod 26	**21**	**7**	**5**	**23**	**20**	**11**	**22**	**1**
Ciphertext Letter	**V**	**H**	**F**	**X**	**U**	**L**	**W**	**B**

Ciphertext = VHFXULWB

Q2. Decrypt the ciphertext KHOOR using Caesar Cipher with key K = 3

Ciphertext Letter	**K**	**H**	**O**	**O**	**R**
Ciphertext Value (C)	**10**	**7**	**14**	**14**	**17**
Key (K)	**3**	**3**	**3**	**3**	**3**
C − K	**7**	**4**	**11**	**11**	**14**
(C − K) mod 26	**7**	**4**	**11**	**11**	**14**
Plaintext Letter	**H**	**E**	**L**	**L**	**O**

Plaintext = HELLO

Q3. Encrypt the plaintext NETWORK using a shift of 7

Plaintext Letter	**N**	**E**	**T**	**W**	**O**	**R**	**K**
Plaintext Value (P)	**13**	**4**	**19**	**22**	**14**	**17**	**10**
Key (K)	**7**	**7**	**7**	**7**	**7**	**7**	**7**
P + K	**20**	**11**	**26**	**29**	**21**	**24**	**17**
(P + K) mod 26	**20**	**11**	**0**	**3**	**21**	**24**	**17**
Ciphertext Letter	**U**	**L**	**A**	**D**	**V**	**Y**	**R**

Ciphertext = ULADVYR

Q4. Decrypt the ciphertext JVTLZAOLY using a shift value of 7

Ciphertext Letter	**J**	**V**	**T**	**L**	**Z**	**A**	**O**	**L**	**Y**
Ciphertext Value (C)	**9**	**21**	**19**	**11**	**25**	**0**	**14**	**11**	**24**
Key (K)	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**
C − K	**2**	**14**	**12**	**4**	**18**	**-7**	**7**	**4**	**17**
(C − K) mod 26	**2**	**14**	**12**	**4**	**18**	**19**	**7**	**4**	**17**
Plaintext Letter	**C**	**O**	**M**	**E**	**S**	**T**	**H**	**E**	**R**

Plaintext = COMESTHER

Q5. If the plaintext letter is P and the key is 11, determine the corresponding ciphertext letter.

Plaintext Letter	**Plaintext Value (P)**	**Key (K)**	**P + K**	**(P + K) mod 26**	**Ciphertext Letter**
P	**15**	**11**	**26**	**0**	**A**

Q6. Using Caesar Cipher, encrypt the word COMPUTER with key 13 (ROT13)

Plaintext Letter	**C**	**O**	**M**	**P**	**U**	**T**	**E**	**R**
Plaintext Value (P)	**2**	**14**	**12**	**15**	**20**	**19**	**4**	**17**
Key (K)	**13**	**13**	**13**	**13**	**13**	**13**	**13**	**13**
P + K	**15**	**27**	**25**	**28**	**33**	**32**	**17**	**30**
(P + K) mod 26	**15**	**1**	**25**	**2**	**7**	**6**	**17**	**4**
Ciphertext Letter	**P**	**B**	**Z**	**C**	**H**	**G**	**R**	**E**

Ciphertext = PBZCHGRE

Q7. A ciphertext QEB NRFZH YOLTK CLU GRJMP LSBO QEB IXWV ALD is encrypted using Caesar Cipher. Determine the plaintext.

This ciphertext is encrypted using a shift of 23 (or decrypt by shifting 3 positions forward).

Ciphertext	**QEB**	**NRFZH**	**YOLTK**	**CLU**	**GRJMP**	**LSBO**	**QEB**	**IXWV**	**ALD**
Plaintext	**THE**	**QUICK**	**BROWN**	**FOX**	**JUMPS**	**OVER**	**THE**	**LAZY**	**DOG**

Plaintext = THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG

Q8. Encrypt the message CYBER ATTACK using Caesar Cipher with key 5 (Ignore spaces)

Plaintext Letter	**C**	**Y**	**B**	**E**	**R**	**A**	**T**	**T**	**A**	**C**	**K**
Plaintext Value (P)	**2**	**24**	**1**	**4**	**17**	**0**	**19**	**19**	**0**	**2**	**10**
Key (K)	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**
P + K	**7**	**29**	**6**	**9**	**22**	**5**	**24**	**24**	**5**	**7**	**15**
(P + K) mod 26	**7**	**3**	**6**	**9**	**22**	**5**	**24**	**24**	**5**	**7**	**15**
Ciphertext Letter	**H**	**D**	**G**	**J**	**W**	**F**	**Y**	**Y**	**F**	**H**	**P**

Ciphertext = HDGJWFYYFHP

Q9. Decrypt the ciphertext MJQQTBTWQI using key 5

Ciphertext Letter	**M**	**J**	**Q**	**Q**	**T**	**B**	**T**	**W**	**Q**	**I**
Ciphertext Value (C)	**12**	**9**	**16**	**16**	**19**	**1**	**19**	**22**	**16**	**8**
Key (K)	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**
C − K	**7**	**4**	**11**	**11**	**14**	**-4**	**14**	**17**	**11**	**3**
(C − K) mod 26	**7**	**4**	**11**	**11**	**14**	**22**	**14**	**17**	**11**	**3**
Plaintext Letter	**H**	**E**	**L**	**L**	**O**	**W**	**O**	**R**	**L**	**D**

Plaintext = HELLOWORLD

Q10. A Caesar Cipher uses the formula $C = (P + K) mod 26$. Encrypt the numerical sequence $P = \{2, 14, 3, 4\}$using $K = 8$.

Plaintext Value (P)	**2**	**14**	**3**	**4**
Key (K)	**8**	**8**	**8**	**8**
P + K	**10**	**22**	**11**	**12**
(P + K) mod 26	**10**	**22**	**11**	**12**
Ciphertext Value (C)	**10**	**22**	**11**	**12**

Encrypted Numerical Sequence = {10, 22, 11, 12}

Q11. Construct the Playfair Matrix using the keyword MONARCHY. Then encrypt the plaintext BALLOON.

Step 1: Construct the Playfair Matrix

- **Keyword: MONARCHY**
- **Remove duplicate letters.**
- **Combine I and J into one cell.**

- **Fill the remaining letters of the alphabet.**
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M	**O**	**N**	**A**	**R**
C	**H**	**Y**	**B**	**D**
E	**F**	**G**	**I/J**	**K**
L	**P**	**Q**	**S**	**T**
U	**V**	**W**	**X**	**Z**

Step 2: Prepare the Plaintext

Plaintext: BALLOON

Split into digraphs:

BA | LX | LO | ON

- **Double LL → insert X**
- **Final pairs: BA | LX | LO | ON**

Step 3: Encrypt Each Pair

Pair	Rule Applied	Cipher Pair
BA	Same column	IB
LX	Rectangle	SU
LO	Rectangle	PM
ON	Same row	NA

Encrypted Ciphertext

IBSUPMNA

Q12. Construct the Playfair Matrix using the keyword SECURITY. Encrypt the plaintext NETWORK.

Step 1: Construct the Playfair Matrix

Keyword: SECURITY (remove duplicate letters)

S	**E**	**C**	**U**	**R**
I/J	**T**	**Y**	**A**	**B**
D	**F**	**G**	**H**	**K**

L	**M**	**N**	**O**	**P**
Q	**V**	**W**	**X**	**Z**

Step 2: Prepare the Plaintext

Plaintext: NETWORK

Split into digraphs:

NE | TW | OR | KX

- **Since K has no pair, add X.**
- **Final pairs: NE | TW | OR | KX**

Step 3: Encrypt Each Pair

Pair	Rule Applied	Cipher Pair
NE	Rectangle	CM
TW	Rectangle	YV
OR	Rectangle	PU
KX	Rectangle	HZ

Encrypted Ciphertext

CMYVPUHZ

Q13. Using the keyword COMPUTER, Encrypt the plaintext INFORMATION

Step 1: Construct the Playfair Matrix

Keyword: COMPUTER (remove duplicate letters)

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I/J
K	L	N	Q	S
V	W	X	Y	Z

Step 2: Prepare the Plaintext

Plaintext: INFORMATION

Split into digraphs:

IN | FO | RM | AT | IO | NX

- **Since the last letter N has no pair, add X.**
- **Final pairs: IN | FO | RM | AT | IO | NX**

Step 3: Encrypt Each Pair

Pair	Rule Applied	Cipher Pair
IN	Rectangle	GS
FO	Rectangle	EL
RM	Same Column	GX
AT	Same Row	BE
IO	Rectangle	FU
NX	Same Column	XR

Encrypted Ciphertext

GSELGXBEFUXR

Q14. Construct the Playfair Matrix using the keyword CRYPTOGRAPHY and decrypt the ciphertext BMNDZBXDKYBEJVDMUIXMMNUVIF

Step 1: Construct the Playfair Matrix

Keyword: CRYPTOGRAPHY (remove duplicate letters, combine I/J)

C	R	Y	P	T
O	G	A	H	B
D	E	F	I/J	K
L	M	N	Q	S
U	V	W	X	Z

Step 2: Split the Ciphertext into Digraphs

BM | ND | ZB | XD | KY | BE | JV | DM | UI | XM | MN | UV | IF

Step 3: Decrypt Each Pair

Cipher Pair	Plain Pair
BM	GS

Cipher Pair	Plain Pair
ND	FL
ZB	TK
XD	UI
KY	FT
BE	KG
JV	EX
DM	LE
UI	DX
XM	QV
MN	NL
UV	VW
IF	FH

Decrypted Plaintext

GSFLT KUIFTKGEXLEDXQVNLVWFH

Answer: GSFLTKUIFTKGEXLEDXQVNLVWFH

Q15. Generate the Playfair Key Matrix using ENGINEERING and Encrypt HELLO WORLD

Step 1: Construct the Playfair Matrix

Keyword: ENGINEERING (remove duplicate letters, combine I/J)

E	N	G	I/J	R
A	B	C	D	F
H	K	L	M	O
P	Q	S	T	U
V	W	X	Y	Z

Step 2: Prepare the Plaintext

Plaintext: HELLO WORLD

Remove spaces:

HELLOWORLD

Split into digraphs:

HE | LX | LO | WO | RL | DX

- **Double LL → insert X**
- **Last D has no pair → add X**

Step 3: Encrypt Each Pair

Pair	Rule Applied	Cipher Pair
HE	Rectangle	OA
LX	Rectangle	SZ
LO	Same Row	MH
WO	Rectangle	ZK
RL	Rectangle	GO
DX	Rectangle	CY

Encrypted Ciphertext

OASZMHZKGOCY